Suvasish Das

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Education

- **Indian Institute of Technology, Gandhinagar** **Gujarat, India**
  PhD in Artificial Intelligence - Ongoing *Jul, 2026 - Jul, 2031*
- **Indian Institute of Technology, Bombay** **Mumbai, India**
  PhD in Intelligent Systems (IS1) - Dropped due to research misalignment *Jan, 2025 – Jul, 2026*
- **West Bengal State University** **Kolkata, India**
  M.Sc (Hons) in Computer Science (**CGPA:** 9.78/10) *Jul 2022 – Jul 2024*
  Secure 1st position in the department and its affiliated colleges - `[Certificate]`
- **Barrackpore Rastraguru Surendranath College** **Kolkata, India**
  B.Sc (Hons) in Computer Science (**CGPA:** 9.51/10) *Jul 2019 – Jul 2022*

Recent Works

Least Unsquared Deviation for Cryo-EM 3D Reconstruction `[Files]` `[Report]` Jul, 2026

- Implementation of Algorithms and Comparative Study of Least Unsquared Deviation.
- Used diffusion for denoising 2D projections, which significantly reduces noise in the input images.

Document Image Machine Translation (ENG to CHI) `[Presentation]` Nov, 2025

- Developed an end-to-end document image translation pipeline that converts English document images into structure-preserving Chinese Markdown, and obtained LLM finetuned structural accuracy of 88.47%

Text-to-text Machine Transliteration (Hindi to Devangari) `[Presentation]` `[Code]` Sep, 2025

- A transliteration system to convert Hindi (Roman) to Devanagari while preserving pronunciation accuracy.
- Experimented with LSTM, Transformer, and SoTA LLM-based architectures and compared them.

Part of Speech Tagging for English `[Presentation]` Aug, 2025

- Developed a POS tagger for the English language using different architectures such as HMM, Bi-RNN, SoTA LLM, etc and compared them.

Diverse and Improved Sparse Retrieval `[Report1]` `[Report2]` Jul, 2025

- Developed a retrieval system to preserve diversity in the generated outputs on the WikiDump corpus.
- Improved the system by adding a learned sparse matrix in the latent space, with added validation such as vocabulary mismatch, synonymy, or cross-lingual variation, etc.

Two Stage Ranking System `[Report]` Sep, 2025

- Stage-1 Ranker module generates relevant documents using a contrastive learning strategy.
- Stage-2 Reranker module leverages Markov random fields to model term dependencies within a window.

Reproducing Conditional Generative Adversarial Networks `[Report]` `[Code]` Nov, 2025

- Reproduce a 50% scaled-down version of the Conditional-GAN paper by [M Mirza and S Osindero].

Graph Neural Network based Question Answering system `[Report]` `[Code]` Apr, 2025

- Developed a Graph Neural Network-based end-to-end system for information retrieval.
- Used the FLAN T5 language model for in-context text generation for user-friendly responses.

Information Bottleneck and Classification using VAE `[Presentation]` Mar, 2025

Jailbreaking and Red-teaming DeepSeek Models [Presentation][Files] Nov, 2025

- Studied adversarial attacks based on prompt injection to break the safeguard of DeepSeek models and compared our implementations with the current available benchmarks.

Bengali Spelling Correction using Noisy Channel Approach [Code] [Report] Jun 2024

- A web-based application to rectify Bengali spelling errors by using Bayesian Inference.
- Implemented beam-search on top of the noisy-channel to improve overall language model accuracy.
- Collected diverse handwritten texts from college students for real-time evaluation.

Facility Detection and Popularity Assessment [Code] [Slides] Mar 2023

- Facility detection from large-scale text classification of social media and crowdsourced data.
- Achieve accuracy of 91% using Support Vector Machines and 85% using Naive Bayes.
- Scrape Twitter for a particular timeline with target facility type, used to determine the truthiness of the tweet.

Non-adaptive Image Encryption based on Chaos [Code] [Report] Jul 2022

- Implemented an image encryption method based on non-adaptiveness chaos.
- Reduced noise up to 80% created by public channels and retrieved encrypted images at the receivers end.

A full featured Technical Blog Website using Python-Flask [Code] 2021

Android based Covid-19 Vaccine Notifier Application [Code] 2020

Coursework / Certifications

Courseworks: Information Retrieval, Natural Language Processing, Foundations of Machine Learning, Advanced Image Processing (Inverse Problem), Human-centered AI, etc.

Online Certifications: Neural Networks and Deep Learning (Coursera), Neural Network Bootcamp (Udemy)

Skills

Languages: Python (8000+ lines), C/C++, HTML/CSS, Java, JavaScript, LaTeX, SQL

Library & Tools: Pytorch, Tensorflow, NLTK, Pandas, Numpy, Keras, Flask, Git, Jekyll

Experience

Teaching Assistant in CS101 (Fall-25), CS271 (Spring-26) IIT Bombay **Mumbai, India**
Assisted UG students by arranging class quizzes, assignments, and semester exams, etc. *Jul, 2025 – Jul, 2026*

Tech Content Reviewer/ Creator in CS101-outreach program by BodhiTree, IIT Bombay **Mumbai, India**
Content creator and reviewer for BodhiTree platform, sponsored by IBM and TowerResearch *Jul, 2025 – Jul, 2026*

Fellowships / Achivements

Received Honorarium *of Rs. 1.2 Lakh from BodhiTree, sponsored by IBM and TowerResearch* *Feb, 2026*

INSPIRE Fellowship (round-1) *awarded by Department of Science and Technology, Govt. of India* *Jun, 2025*

GATE Qualified *with a score of 469 and 94.28% pencentile (paper: CS), 76.56% percentile (paper: DA)* *Mar, 2024*

UGC-NET Qualified $\times 2$ *for assistant professor and admission to PhD* *Dec, 2022 & Jun, 2024*